Figure 1

		Status quo	Shift	Example actionable items
Mindset & Skills	1	Siloed disciplines, insufficient interdisciplinary training	Cultivating a multidisciplinary mindset	Leverage LLMs for self-directed AI training, Early course embedded exposure to interdisciplinary topics
	2	Vibe approaches for coding & hypothesis generation with low understanding of underlying syntax and logic	Knowledge-based thinking	Learn basic coding, Develop a strong foundation in subject knowledge, Build resiliency to rapidly changing tool sets
Classroom practice	3	Restricting or banning GenAI in classrooms	Embracing GenAI in plant science classroom	Process-based assessment, Explicit AI integration in coursework and student projects
	4	Unskeptical use of GenAI responses	Mitigating biases and hallucinations inherent in plant science applications	Discuss GenAI biases in classroom, Develop critical thinking skills, Validate output
Ethics & Infrastructure	5	Insufficient awareness of ethical issues surrounding GenAI use	Encouraging ethical GenAI use in plant science applications	Incorporate AI ethics, misconduct, data security and sovereignty issues in classroom teaching
	6	Relying on fragmented infrastructure, training resources and coursework	Developing global infrastructures for GenAI pedagogy in plant sciences	Empower core infrastructure, Develop global plant science training resources and datasets

Figure 1: A roadmap to GenAI literacy in plant sciences. Current challenges include siloed disciplines, uncritical vibe-coding, reliance on biased data and insufficient AI knowledge. Six strategic shifts are proposed to transform plant science education & research, moving from passive use to active, knowledge-based, ethical engagement. The outcome is a global, interconnected community where GenAI complements disciplinary knowledge to solve complex plant science problems.